

Planning and Reinforcement Learning in AI Systems

(PARLAI)

Lesson 10b - Accounting for Risk and Model-Uncertainty

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Acknowledgements

IJCAI 2024 Tutorial on Safe Reinforcement Learning Ming Jin and Shangding Gu

https://people.eecs.berkeley.edu/~shangding.gu/tutorials/Safe_RL_IJCAI_Tutorial_2024.pdf

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Stochastic fully observable environments

Stochastic fully observable environments are defined using a **Markov Decision Process (MDP)** $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ where

- $\mathcal{S}$ is a finite set of states
- $\mathcal{A}$ is a finite set of actions
- $\mathcal{P}$ is a state transition probability matrix
$$\mathcal{P}_{s,s'}^a = \mathcal{P}[s_{t+1} = s' | s_t = s, t = a]$$
- $\mathcal{R}$ is a reward function, $\mathcal{R}_s^A = \mathbb{E}[R_{t+1} | s_t = s, t = a]$, and
- **optional:** γ is a discount factor $\gamma \in [0, 1]$ that is used to favor immediate rewards over future rewards.

The Markov property: “The future is independent of the past given the present”.

Extensions: Infinite and continuous MDPs, partially observable MDPs, undiscounted, average reward MDPs. etc.

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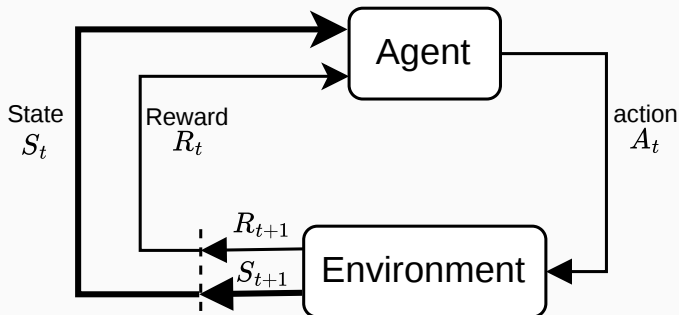
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Stochastic Fully Observable Environments



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Stochastic Fully Observable Environments

A solution to an MDP is a **policy** that prescribes the action to be taken at every state.

- **Deterministic Policy:** a mapping $\pi : \mathcal{S} \rightarrow \mathcal{A}$ from states to actions.
- **Stochastic Policy:** a mapping $\pi : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$ from state and action pairs to the probability $\pi(a|s)$ of taking action a when in state s .

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Motivation for Reliable Decision-Making

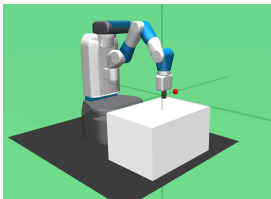
Motivation

So far, we have considered agents that aim to maximize their utility while exploring the environment.

The main concern was how to balance exploration and exploitation

We also accounted for the issue of partial and imperfect knowledge.

But the core assumption so far was that the world is safe to explore.



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Motivation

In many domains, exploration is costly and maximizing expected utility is not appropriate.



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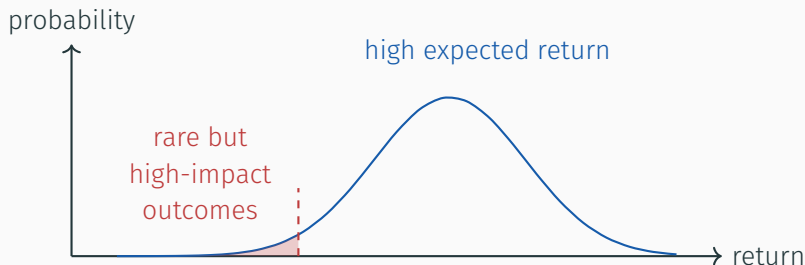
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The Gap: Expected Return Is Not Enough

Most standard RL (and multi-agent RL) algorithms optimize expected cumulative reward.



In high-stakes systems, a policy can have good average performance while still producing unacceptable failures.

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Approaches to Reliability

Safe RL, Robust RL, and Related Settings

Different extensions ask for reliability in different ways.

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Safe RL, Robust RL, and Related Settings

Different extensions ask for reliability in different ways.

Safe RL focuses on constraint violations.

Can the agent learn and act without violating safety constraints?

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Safe RL, Robust RL, and Related Settings

Different extensions ask for reliability in different ways.

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Can the agent learn and act without violating safety constraints?

Robust RL and Bayesian RL focus on uncertainty in the environment.

Does the policy still work under model errors, disturbances, or distribution shift?

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Safe RL, Robust RL, and Related Settings

Different extensions ask for reliability in different ways.

Safe RL focuses on constraint violations.

Can the agent learn and act without violating safety constraints?

Robust RL and Bayesian RL focus on uncertainty in the environment.

Does the policy still work under model errors, disturbances, or distribution shift?

Risk-sensitive RL focuses on the tail of the outcome distribution.
How bad is performance in the worst cases?

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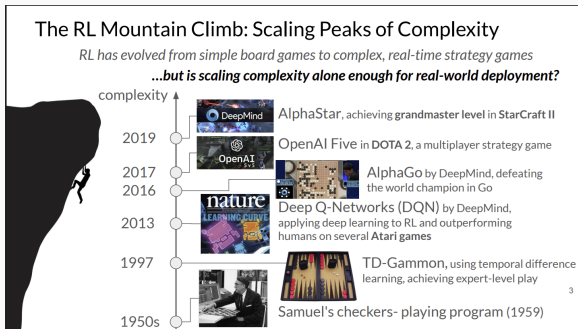
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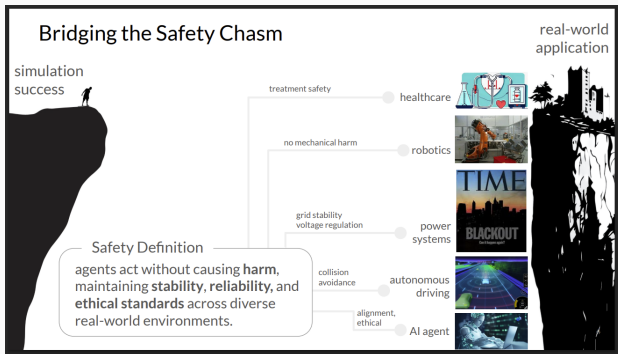
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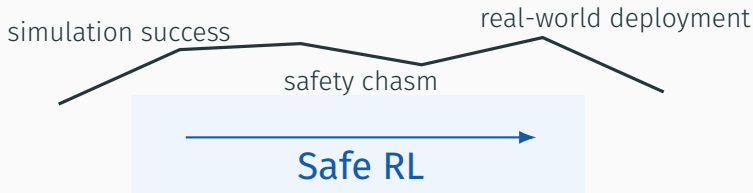
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Why Safe RL?

Safe RL studies how learning agents can improve performance while avoiding unacceptable behavior.



Safe RL connects standard reward optimization with constraints, verification, runtime assurance, and trustworthy deployment.

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Why Safe RL?

Safe RL bridges standard RL to deployable, trustworthy autonomous systems.

Core question

How can an RL agent improve performance while avoiding unacceptable behavior during learning and deployment?

- Safe RL formulations: CMDPs and safety constraints.
- Model-free safe RL: primal-dual and primal methods.
- Model-based safe RL: safety certificates and uncertainty.
- Multi-agent safe RL: safety under interaction.
- Benchmarks, verification, LLM safety, and future directions.

Safe RL is not one algorithm; it is a family of methods for optimizing reward while explicitly reasoning about safety.

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What Does “Safe” Mean?

Different safety requirements lead to different mathematical formulations.

- **State safety:** avoid states that violate physical, legal, or operational limits.
- **Action safety:** filter actions that exceed actuator limits or enter forbidden control regimes.
- **Trajectory safety:** constrain cumulative cost, probability of failure, or worst-case outcome.
- **System safety:** guarantee recoverability, stability, interpretability, and auditability.

Two important axes

- **When?** during learning $\leftrightarrow$ only after convergence.
- **How strict?** hard constraints $\leftrightarrow$ soft penalties / risk preferences.

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Safe Reinforcement Learning

Safe RL develops agents that learn effective policies while satisfying safety constraints during learning and deployment.

Typical mechanisms:

- constrained policy optimization,
- Lyapunov-based guarantees,
- formal shielding,
- recovery strategies.

We will only explore a subset of these.

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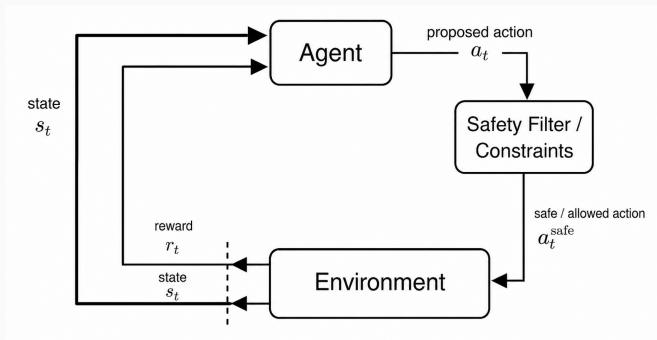
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Types of Safety Constraints

Different safety requirements lead to different types of constraints.

Cumulative constraints $\mathbb{E}_{\pi} [\sum_{t=0}^{\infty} \gamma^t C(s_t, t)] \leq d$

Useful when safety violations accumulate over time.

Instantaneous constraints $\mathbb{E}_{\pi}[C(s_t, t)] \leq d_t$

Useful when each individual step must remain safe.

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Useful when safety violations accumulate over time.

Instantaneous constraints $\mathbb{E}_{\pi}[C(s_t, t)] \leq d_t$

Useful when each individual step must remain safe.

Another axis

- **Expectation-based:** safety holds on average.
- **Probability-based:** probability of violation must be small.

Choosing the right constraint depends on whether the domain tolerates rare failures.

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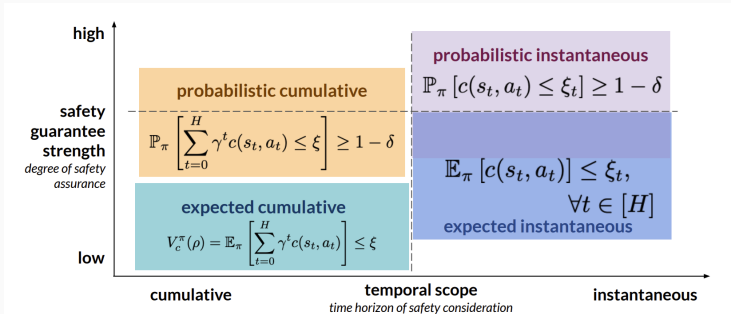
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Performance Objective

In RL, V^π and $J(\pi)$ describe related but different objects.

The **value function** depends on where the agent starts:

$$V^\pi(s) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, t) \mid s_0 = s \right].$$

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Performance Objective

In RL, V^π and $J(\pi)$ describe related but different objects.

The **value function** depends on where the agent starts:

$$V^\pi(s) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, t) \mid s_0 = s \right].$$

The **performance objective** averages over the initial-state distribution ρ_0 :

$$J(\pi) = \mathbb{E}_{s_0 \sim \rho_0} [V^\pi(s_0)].$$

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Safe RL as a Constrained MDP

A common formulation of safe RL uses a **Constrained Markov Decision Process** (CMDP), which extends an MDP with cost functions and corresponding safety budgets:

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \rho_0, \mathcal{R}, \gamma, \{C_i\}_{i=1}^m, \{d_i\}_{i=1}^m).$$

- $\mathcal{S}$: state space.
- $\mathcal{A}$: action space.
- $\mathcal{P}(s' | s, a)$: probability of transitioning to state s' after taking action a in state s .
- $\rho_0(s)$: distribution over the initial state s_0 .
- $\mathcal{R}(s, a)$: task reward obtained by taking action a in state s .
- $\gamma \in [0, 1)$: discount factor.
- $C_i(s, a)$: the i -th cost function, measuring a particular type of unsafe or undesirable behavior.
- d_i : maximum allowed cumulative value, or **budget**, for cost C_i .
- m : number of safety constraints.

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Safe RL as a Constrained Optimization Problem

The objective is to maximize the expected discounted reward:

$$J_R(\pi) = \mathbb{E}_{\substack{s_0 \sim \rho_0 \\ t \sim \pi(\cdot | s_t) \\ s_{t+1} \sim P(\cdot | s_t, t)}} \left[\sum_{t=0}^{\infty} \gamma^t \mathcal{R}(s_t, t) \right].$$

For every safety requirement i , the expected discounted cost must remain below its allowed budget:

$$J_{C_i}(\pi) = \mathbb{E}_{\pi, \mathcal{P}, \rho_0} \left[\sum_{t=0}^{\infty} \gamma^t C_i(s_t, t) \right] \leq d_i, \quad i = 1, \dots, m.$$

$$\begin{array}{ll} \max_{\pi} & J_R(\pi) \\ \text{subject to} & J_{C_i}(\pi) \leq d_i, \quad i = 1, \dots, m \end{array}$$

The reward describes what the agent should achieve, while the costs and budgets specify which behaviors are acceptable during learning and execution.

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Safe RL as a Constrained MDP

$$\max_{\pi} J_R(\pi) \quad \text{s.t.} \quad J_{C_i}(\pi) \leq d_i, \quad i = 1, \dots, m$$

A CMDP augments the usual reward objective with explicit cost constraints.

Interpretation:

- **Reward:** what the agent should achieve.
- **Cost:** what the agent should avoid.
- **Budget:** tolerated level of violation.

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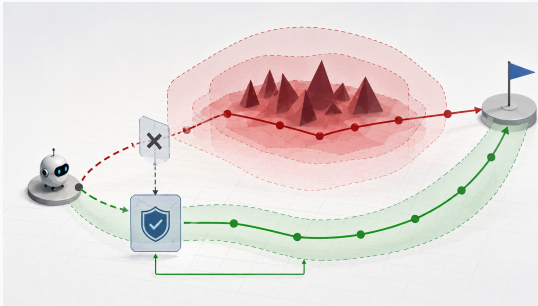
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- Reward: reach the goal quickly.
- Cost: enter danger zone, collide, exceed speed.
- Budget: collision rate below a threshold.



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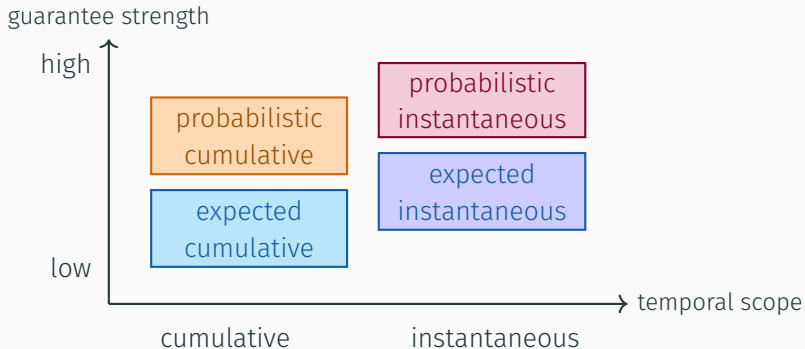
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Constraint Types: Main Tradeoff



Stronger safety guarantees typically increase computational difficulty and conservatism.

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Instantaneous and State-Based Constraints

Some systems require safety at each step, not only over the whole episode.

Expected instantaneous constraint

$$\mathbb{E}_{\pi} [C(s_t, t)] \leq d_t, \quad \forall t.$$

Provides immediate safety feedback.

Probabilistic instantaneous: state constraint

$$\Pr_{\pi}(s_t \in s_{\text{safe}}) \approx 1, \quad \forall t.$$

Focuses directly on avoiding unsafe states.

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Instantaneous and State-Based Constraints

Expectation constraints control average cost. Probability constraints control the chance of violation.

Probability-based constraints are closer to operational safety requirements, but they are usually more difficult to optimize.

Tradeoff: Instantaneous constraints give stronger local control over violations, but may be more conservative and harder to satisfy in stochastic systems.

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Probability-Based Safety Constraints

Chance constraint on cumulative cost

$$\Pr_{\pi} \left(\sum_{t=0}^H \gamma^t C(s_t, t) \leq \xi \right) \geq 1 - \delta.$$

Controls the probability that a full trajectory exceeds the safety budget.

Chance constraint on immediate cost

$$\Pr_{\pi} (C(s_t, t) \leq \xi_t) \geq 1 - \delta, \quad \forall t.$$

Controls violations at each decision step.

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Using Unconstrained RL for a CMDP

Can we use the methods we have seen so far for standard RL?

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Using Unconstrained RL for a CMDP

Can we use the methods we have seen so far for standard RL?
The most common approach is to limit expected cumulative cost:

$$J_C(\pi) = \mathbb{E}_{\pi} \left[\sum_{t=0}^H \gamma^t C(s_t, t) \right] \leq d.$$

If we want to use an RL algorithm that only supports maximizing expected return, we introduce a nonnegative multiplier (λ):

$$J_R(\pi) \lambda (J_C(\pi) - d).$$

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Using Unconstrained RL for a CMDP

For a fixed (λ), this is an ordinary RL objective with a modified reward.

Therefore, any standard RL algorithm can be used to optimize

$$\max_{\pi} \mathbb{E} \pi \left[\sum_{t=0}^H \gamma^t (\mathcal{R}(s_t, t) - \lambda C(s_t, t)) \right].$$

The constrained problem becomes an ordinary RL problem with an adaptively chosen cost penalty.

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Using Unconstrained RL for a CMDP

$$\max_{\pi} \mathbb{E} \pi \left[\sum_{t=0}^H \gamma^t (\mathcal{R}(s_t, t) - \lambda C(s_t, t)) \right].$$

Why it is useful

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Using Unconstrained RL for a CMDP

$$\max_{\pi} \mathbb{E} \pi \left[\sum_{t=0}^H \gamma^t (\mathcal{R}(s_t, t) - \lambda C(s_t, t)) \right].$$

Why it is useful

- Mirrors the reward objective.
- Works naturally with policy-gradient methods.
- Supports long-horizon resource budgets.

Main limitation

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Using Unconstrained RL for a CMDP

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Why it is useful

- Mirrors the reward objective.
- Works naturally with policy-gradient methods.
- Supports long-horizon resource budgets.

Main limitation

- Safety is enforced only on average.
- Rare dangerous trajectories can be hidden.
- Feedback about violations may be delayed.

Expected cumulative constraints are easy to optimize, but they do not guarantee that every trajectory is safe.

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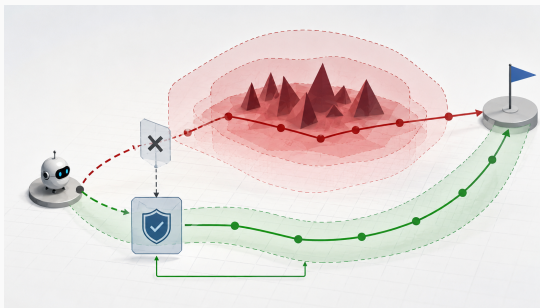
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How to set the costs appropriately?



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Solving CMDPs: Lagrangian Relaxation

A more rigorous approach requires finding the right Lagrange multipliers $\lambda_i \geq 0$.

This becomes an optimization problem.

$$\max_{\pi} \min_{\lambda \geq 0} \mathbb{E}^{\pi} \left[\sum_{t=0}^{\infty} \gamma^t \left(R(s_t, t) - \sum_{i=1}^m \lambda_i C_i(s_t, t) \right) \right] + \sum_{i=1}^m \lambda_i d_i.$$

Interpretation

- If a constraint is violated, its multiplier increases.
- The policy is discouraged from choosing actions that generate high cost.
- The resulting objective can be optimized with policy-gradient or actor-critic methods.

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Primal-Dual Safe Policy Optimization

For one constraint, the Lagrangian objective is:

$$L(\pi, \lambda) = J_R(\pi) - \lambda(J_C(\pi) - d), \quad \lambda \geq 0.$$

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Primal-Dual Safe Policy Optimization

For one constraint, the Lagrangian objective is:

$$L(\pi, \lambda) = J_R(\pi) - \lambda(J_C(\pi) - d), \quad \lambda \geq 0.$$

Policy update

$$\pi \leftarrow \pi + \alpha \nabla_{\pi} L(\pi, \lambda)$$

Improves reward while accounting for safety penalty.

Multiplier update

$$\lambda \leftarrow [\lambda + \beta(J_C(\pi) - d)]_+$$

Increases the safety penalty when the constraint is violated.

The multiplier controls the balance between task reward and constraint satisfaction.

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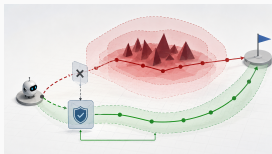
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Why Reward and Penalties Are Not Enough

Instead of keeping reward and safety separate, we define a shaped reward:

$$\tilde{\mathcal{R}}(s, a, s') = \mathcal{R}(s, a, s') - \lambda C(s, a, s'),$$

where $C(s, a, s')$ measures safety violation and λ controls how strongly violations are penalized.



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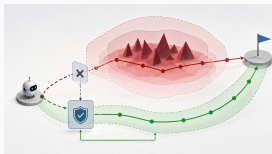
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Penalty methods hide the safety tradeoff inside a scalar reward.

For safety-critical domains, we may want a separate signal: reward for task performance, cost for violations, and constraints for acceptable risk.

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Why Reward and Penalties Are Not Enough

Encoding safety as a negative reward often changes behavior, but does not provide a guarantee.

Penalty shaping

$$\max_{\pi} \mathbb{E}_{\pi} \left[\sum_t \gamma^t (\mathcal{R}(s_t, t) - \lambda C(s_t, t)) \right]$$

A large λ discourages violations, but the agent may still choose unsafe actions when reward is high.

Constraint reasoning

$$\max_{\pi} J_R(\pi) \quad \text{s.t.} \quad J_C(\pi) \leq d$$

The constraint remains visible to the algorithm and to evaluation.

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Model-free and Model-based Safe RL

Model-Free Safe RL learns a policy directly from interaction.

- Does not require an explicit dynamics model.
- Usually optimizes reward subject to cost constraints.
- Examples: Lagrangian methods, CPO, primal-dual policy gradients.

Model-Based Safe RL Uses a [learned] model of the environment to reason about future outcomes.

- Can predict unsafe future states before executing actions.
- Examples: safe MPC, shielding, reachability-based safety filters.

Model-free methods learn safe behavior from experience while model-based methods try to anticipate unsafe outcomes before acting.

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Example Model-Free Approach: Constrained Policy Optimization

Constrained Policy Optimization (CPO) (Achiam et al. 2017) updates the policy while trying to keep safety constraints satisfied.

The CMDP objective is:

$$\max_{\pi} J_R(\pi) \quad \text{s.t.} \quad J_{C_i}(\pi) \leq d_i.$$

CPO performs a policy update of the form:

$$\pi_{\text{new}} = \pi_{\text{old}} + \Delta\pi,$$

but restricts the update so that:

$$J_{C_i}(\pi_{\text{new}}) \leq d_i$$

and the new policy stays close to the old one.

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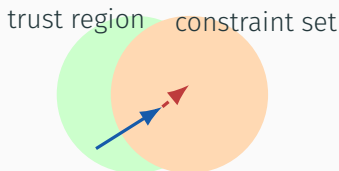
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Example Model-Free Approach: Constrained Policy Optimization

Constrained Policy Optimization tries to improve reward while keeping the new policy close and feasible.

$$\begin{aligned} \max_{\theta} \quad & \text{reward surrogate}(\theta) \\ \text{s.t.} \quad & \text{cost surrogate}(\theta) \leq d, \\ & D_{\text{KL}}(\pi_{\theta_{old}} \parallel \pi_{\theta}) \leq \epsilon. \end{aligned}$$



reward improvement followed by feasibility correction

CPO is model-free: it does not plan with a learned dynamics model. Instead, it uses sampled trajectories to estimate reward, cost, and safe policy updates.

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Example Model-Based Approach: Safe MPC

A model-based safe RL method can use a learned dynamics model:

$$s_{t+1} = \hat{f}(s_t, a_t)$$

to predict the consequences of candidate actions.

At each time step, the agent solves a constrained planning problem:

$$a_t, \dots, a_{t+H} = \arg \max \sum_{k=0}^H \gamma^k \mathcal{R}(s_{t+k}, a_{t+k})$$

s.t. safety constraints: $C_i(s_{t+k}, a_{t+k}) \leq d_i$, $k = 0, \dots, H$ Then it executes only the first action a_t , observes the next state, and replans.

Safe MPC uses the model to look ahead and reject actions that may lead to unsafe future states.

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How to Evaluate Safe RL

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How to Evaluate Safe RL

Performance metrics

- average return, success rate, sample efficiency

Safety metrics

- constraint cost, violation probability, worst episode / CVaR

Deployment metrics

- learning-time violations and intervention rate, recoverability, robustness under perturbations

**Report task performance and safety performance separately.
Do not average away safety failures: show violation curves, not
only final return.**

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Common benchmark families

- AI Safety Gridworlds
- Safety Gym / Safety Gymnasium
- Safe Control Gym
- Safe MuJoCo
- Safe multi-agent MuJoCo, Robosuite, Isaac Gym

A safe RL method should not be evaluated only by final return; violation curves during learning are equally important.

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Takeaways and Open Questions

Safe RL is a family of formulations, not a single algorithm.

Key takeaways

- 1 Keep reward and cost separate.
- 2 Decide whether safety is needed during learning or only after training.
- 3 Match the method to the guarantee you need.
- 4 Evaluate failures explicitly, not only return.

Open questions

- How to scale formal guarantees to deep policies?
- How to handle unknown constraints?
- How to combine foundation models with certifiable safety?
- How safe is safe enough for deployment?

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Risk-Sensitive RL

Standard RL typically optimizes the **expected return**:

$$J(\pi) = \mathbb{E}_{\pi} [G], \quad G = \sum_{t=0}^H \gamma^t r_t.$$

However, two policies can have the same expected return while having very different probabilities of poor outcomes.

Risk-sensitive RL therefore considers the entire distribution of the return, rather than only its mean.

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Risk-Sensitive RL may prefer policies that:

- have lower variability,
- avoid low-return tail events,
- perform well under unfavorable outcomes,
- trade some expected performance for greater reliability.

Risk-sensitive RL asks not only “How well does the policy perform on average?” but also “How bad can its unfavorable outcomes be?”

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Safe vs. Risk-Sensitive RL

Both approaches address undesirable outcomes, but they do so in different ways.



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Risk-Sensitive RL

Optimizes a measure that accounts for the distribution of outcomes:

$$\max_{\pi} \rho(G_R),$$

where ρ may represent CVaR, variance, worst-case return, or a risk-sensitive utility function.

- Expresses a **preference** for reliable outcomes.
- Poor outcomes may still be allowed.
- Risk is part of the objective.
- Usually involves a performance–risk tradeoff.

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Optimizes task performance subject to explicit safety requirements:

$$\max_{\pi} J_R(\pi)$$

$$\text{subject to} \quad J_C(\pi) \leq d.$$

- Defines which behavior is **acceptable**.
- Uses costs, constraints, or action restrictions.
- Safety is separated from task reward.
- Reward should not compensate for severe violations.

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Safe vs. Risk-Sensitive RL

Risk-sensitive RL asks which uncertain outcomes we prefer; safe RL asks which behaviors or outcomes we permit. The two approaches can be combined.

Risk Quantifies uncertainty about unfavorable outcomes, often using the distribution or tail of G_R or G_C :

$$\rho(G_R) \quad \text{or} \quad \rho(G_C).$$

Safety A property or requirement stating that unacceptable behavior should be prevented or sufficiently limited.

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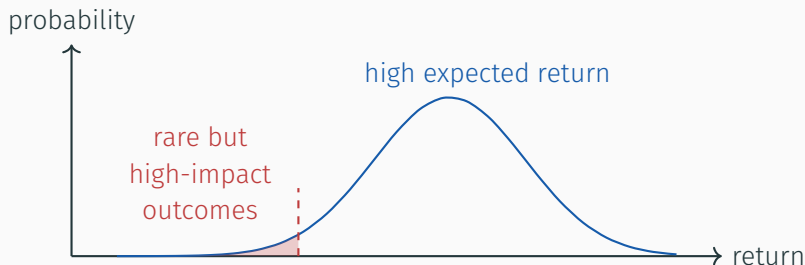
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The Gap: Expected Return Is Not Enough

Most standard RL and multi-agent RL algorithms optimize expected cumulative reward.



In high-stakes systems, a policy can have good average performance while still producing unacceptable failures.

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Value-at-Risk (VaR)

Rather than optimizing $\mathbb{E}[G]$, we may ask:

How bad can the outcome be?

For confidence level $\alpha \in (0, 1)$, the **Value-at-Risk** is the α -quantile of the return:

$$\text{VaR}_\alpha(G) = \inf \{z : P(G \leq z) \geq \alpha\}.$$

Interpretation:

- with probability α , the return is no larger than VaR_α ;
- it represents the threshold separating the worst α fraction of outcomes.

VaR identifies where the worst tail begins, but ignores how bad the tail actually is.

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Conditional Value-at-Risk (CVaR)

CVaR focuses on the entire worst-case tail.

Instead of considering only the quantile, it averages the returns below the VaR threshold.

$$\text{CVaR}_\alpha(G) = \mathbb{E}[G \mid G \leq \text{VaR}_\alpha(G)] .$$

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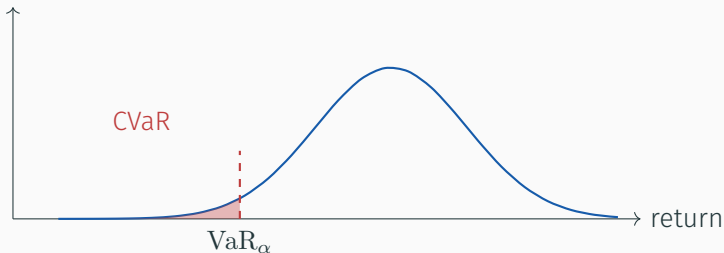
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Conditional Value-at-Risk (CVaR)

CVaR focuses on the entire worst-case tail.

Instead of considering only the quantile, it averages the returns below the VaR threshold.

$$\text{CVaR}_\alpha(G) = \mathbb{E}[G \mid G \leq \text{VaR}_\alpha(G)].$$



CVaR measures the expected performance in the worst α fraction of executions.

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A Bellman Equation for CVaR

Unlike many risk measures,

CVaR admits a Bellman recursion.

For a fixed confidence level α ,

$$V_{\alpha}(s) = \max_a \text{CVaR}_{\alpha}(\mathcal{R}(s, a)) + \gamma V_{\alpha}(S').$$

More generally, CVaR satisfies a dynamic programming recursion in an *augmented state* that includes the confidence level: (s, α) .

The confidence level evolves during the Bellman backup, leading to a time-consistent risk measure.

This allows value iteration, policy iteration, and Q-learning to be generalized to optimize CVaR instead of expected return.

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Theorem: Bellman Principle for CVaR

Theorem (Chow et al., 2015).

For finite discounted MDPs, the CVaR objective satisfies a Bellman recursion over an augmented state space (s, α) .

Consequently,

- value iteration converges,
- policy iteration converges,
- dynamic programming remains applicable.

Key Insight

Unlike many coherent risk measures, CVaR is time consistent when the confidence level is propagated through the Bellman backup.

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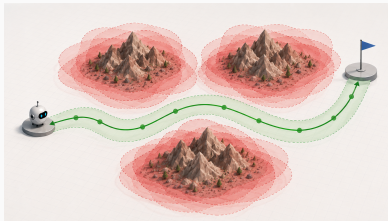
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Robust Reinforcement Learning



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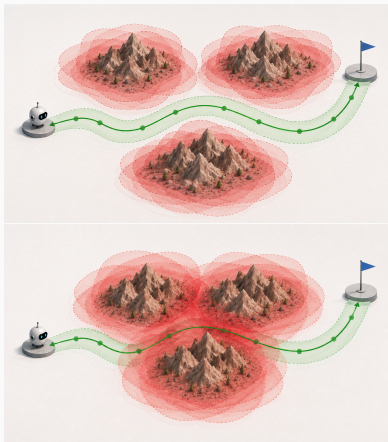
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Robust Reinforcement Learning

Robust RL complements Safe RL by focusing on model uncertainty, disturbances, and adversarial conditions.

Instead of optimizing only for a nominal environment:

$$\max_{\pi} J_M(\pi),$$

robust approaches optimize across a set of possible environments:

$$\max_{\pi} \min_{M \in \mathcal{M}} J_M(\pi).$$

Common methods: robust MDPs, distributionally robust optimization, adversarial training, ensemble-based uncertainty modeling.

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Robust Reinforcement Learning

Robust RL often starts from the idea that the true transition model is not known exactly.

Instead of assuming a single model $\mathcal{P}$, we define an **uncertainty set** $\mathcal{U}_{s,a}$ of plausible transition distributions for each state-action pair.

A typical approach is to optimize against the worst case, assuming an **adversary** selects the least favorable transition model from the uncertainty set.

$$V(s) = \max_{a \in \mathcal{A}} \min_{\mathcal{P}(\cdot | s, a) \in \mathcal{U}_{s,a}} \left[\mathcal{R}(s, a) + \gamma \sum_{s'} P(s' | s, a) V(s') \right].$$

Robust MDPs turn model uncertainty into a max-min Bellman backup. The price is conservatism: the policy may sacrifice nominal performance for reliability.

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The true environment dynamics are often unknown. Instead of assuming a single transition model, robust RL considers a set of plausible models: $\mathcal{P}(\cdot \mid s, a) \in \mathcal{U}(s, a)$, where $\mathcal{U}(s, a)$ is the **uncertainty set** for state-action pair (s, a) .

Possible constructions

- confidence intervals estimated from data,
- bounded ranges of physical parameters,
- models generated by an ensemble,
- distributions within a KL or Wasserstein distance.

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Robust decision rule

The agent evaluates each action using its least favorable plausible model:

$$Q(s, a) = \min_{P \in \mathcal{U}(s, a)} [r(s, a) + \gamma \sum_{s'} P(s' | s, a) V(s')].$$

It then selects:

$$\pi(s) \in \arg \max_a Q(s, a).$$

A larger uncertainty set provides protection against more model errors, but usually produces a more conservative policy.

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Uncertainty Sets

An uncertainty set represents the plausible transition models or environment parameters.

Data-driven uncertainty

- **Confidence regions:** account for finite transition data.
- **Model ensembles:** disagreement between learned models.
- **Distributional neighborhoods:** include models close to a nominal estimate under KL or other measures.

Knowledge-driven uncertainty

- **Physical ranges:** uncertain mass, friction, demand, or wind.
- **Engineering tolerances:** bounded sensor, actuator, or forecast errors.
- **Expert assumptions:** plausible disturbances or adversarial changes.

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Different robust RL methods differ in how they define and train against uncertainty.

Training-time mechanisms

- **Domain randomization:** train across many environment parameters.
- **Adversarial training:** let an adversary perturb states, actions, or dynamics.
- **Ensembles:** use disagreement to identify unreliable predictions.

Robust RL is most useful when the deployment environment may differ from training: forecast errors, disturbances, model misspecification, or distribution shift.

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Example Algorithm: Adversarial Robust Policy Training

A concrete robust RL approach trains the policy against an adversary that selects unfavorable disturbances or environment parameters.

$$\max_{\pi_{\theta}} \min_{\xi \in \Xi} J(\pi_{\theta}, \xi),$$

where ξ may represent changes in dynamics, observation noise, external disturbances, or uncertain demand.

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Example Algorithm: Adversarial Robust Policy Training

Training procedure

- 1 Sample an initial environment configuration.
- 2 The adversary selects a disturbance
$$\xi_k \approx \arg \min_{\xi \in \Xi} J(\pi_{\theta_k}, \xi).$$
- 3 Execute π_{θ_k} in the disturbed environment and collect trajectories.
- 4 Update the policy using the collected experience:
$$\theta_{k+1} = \theta_k + \eta \nabla_{\theta} \widehat{J}(\pi_{\theta_k}, \xi_k).$$
- 5 Repeat with new disturbances.

The adversary exposes weaknesses in the current policy, while the policy learns behavior that remains effective under increasingly difficult conditions.

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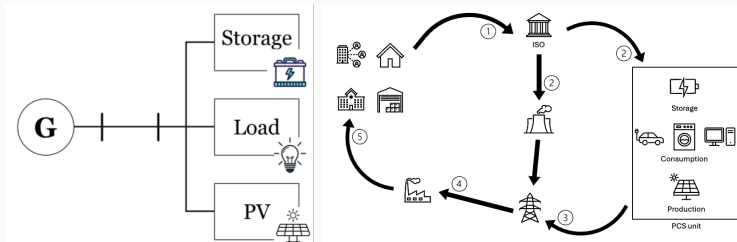
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Energy Systems



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Why [multi-agent] RL for Energy Systems?

Many distributed and strategically autonomous agents:

- grid-edge devices,
- storage units,
- prosumers,
- utility companies,
- system operators,
- market participants.

RL and multi-agent RL provide a natural language for adaptive decision-making in decentralized energy networks.

- adapt to changing prices and conditions,
- coordinate under uncertainty,
- model cooperative and competitive behavior,
- explore decentralized strategies.

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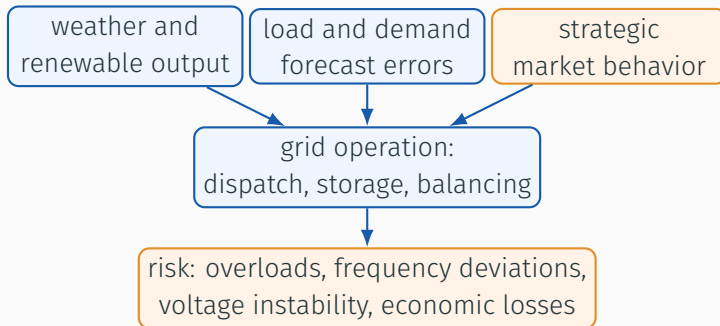
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Energy Systems: Where These Methods Are Needed

In energy systems, uncertainty appears at both the physical and strategic levels.



Risk-aware learning is useful when decisions must adapt to uncertainty without violating operational constraints.

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Safe RL and Robust RL: Complementary Roles

In safe RL: **do not violate constraints.**

- line limits,
- voltage and frequency bounds,
- state-of-charge limits,
- reserve requirements.

In robust RL: **remain reliable under uncertainty.**

- forecast errors,
- renewable variability,
- demand shifts,
- model misspecification.

Safe RL focuses on constraint satisfaction; Robust RL focuses on resilience to uncertainty. Energy systems need both.

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Risk-Aware AI for Energy Systems

Accounting for risk and uncertainty is essential when AI systems operate in **safety-critical**, **dynamic**, and **high-consequence** domains.

Uncertainty

Renewable generation, demand, prices, and forecasts vary over time.

Constraints

Power grids must satisfy physical, operational, and regulatory limits.

Consequences

Unsafe actions may cause overloads, instability, blackouts, or market losses.

The goal is not only to optimize performance, but to make decisions that remain reliable under uncertainty.

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Accounting for Risk and Uncertainty

A wide range of approaches have been developed for decision-making under uncertainty.

Optimization and planning

- **Risk-sensitive optimization:** CVaR, chance constraints.
- **Probabilistic planning:** POMDPs and belief-space planning.
- **Robust control:** reliable behavior across model uncertainty.

Learning-based methods

- **Safe RL:** learn while respecting safety constraints.
- **Robust RL:** learn policies stable under uncertainty.
- **Multi-agent RL:** model decentralized strategic behavior.

These methods provide complementary tools for AI systems that must operate safely and reliably in uncertain environments.

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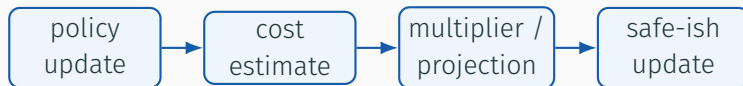
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Algorithm Family 1: Constrained Policy Optimization

Policy-gradient methods can be extended to optimize reward while controlling expected cost.



$$L(\pi, \lambda) = J_{\text{reward}}(\pi) - \lambda(J_{\text{cost}}(\pi) - d).$$

- **Lagrangian methods** adjust λ to penalize constraint violations.
- **CPO-style methods** restrict each update so the new policy stays close and nearly feasible.
- **Main limitation:** expected constraints may still allow rare catastrophic events.

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Algorithm Family 2: Risk-Sensitive and Robust RL

When rare failures matter, the objective must look beyond average return.

Risk-sensitive RL

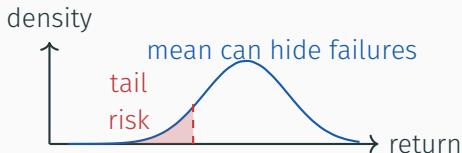
Optimize lower-tail outcomes such as CVaR:

How bad is performance in the worst α fraction of cases?

Robust RL

Optimize against model uncertainty or adversarial disturbances:

Will the policy work if the dynamics are slightly wrong?



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Algorithm Family 3: Shielding and Runtime Assurance

A safety layer can intercept unsafe actions before they reach the environment.

Basic idea

- The RL policy proposes an action.
- A shield checks whether it can lead to violation.
- Unsafe actions are blocked, corrected, or replaced by a backup controller.

It buys strong deployment-time protection when the safety model is correct and a clear separation between learning objective and enforcement.

However, it requires an accurate model of the dynamics and safety constraints, it may block useful actions and it requires online verification

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Algorithm Family 4: Model-Based Safe Exploration

Use a model of dynamics and uncertainty to decide what can be tried safely.

Learn dynamics

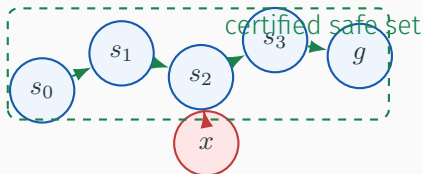
Estimate transition model and epistemic uncertainty from data.

Maintain safe set

Keep states/actions from which the system can recover.

Expand carefully

Explore only when uncertainty bounds still certify safety.



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Takeaways: Main Ideas

- Expected return can hide rare but unacceptable failures.
- **Safe RL** separates reward from safety cost and enforces explicit constraints.
- **CMDPs** are the standard starting point, but the constraint type matters: cumulative, instantaneous, expectation-based, or probability-based.
- **Model-free safe RL** methods such as Lagrangian updates and CPO are flexible, but safety can be sample-intensive.
- **Model-based safe RL** methods such as safe MPC, shielding, and certificates can anticipate unsafe outcomes, but depend on model accuracy.

Reliable RL requires tracking task performance and unacceptable behavior separately.

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Risk-Sensitive RL

Robust RL

Examples

Conclusion

Takeaways: Risk, Robustness, and Applications

- **Risk-sensitive RL** optimizes properties of the return distribution rather than only its mean.
- **VaR** identifies where the bad tail begins; **CVaR** averages the worst tail and admits dynamic programming through an augmented state (s, α) .
- **Robust RL** optimizes performance across model uncertainty, disturbances, adversarial perturbations, and distribution shift.
- **Evaluation** should report reward, cost, violation probability, worst episodes / CVaR, intervention rate, and robustness under perturbations.

Safe RL asks what behavior is allowed; risk-sensitive RL asks how bad the tail is; robust RL asks whether the policy still works when the world changes.

Reminder

Motivation for
Reliable
Decision-Making

Approaches to
Reliability

Safe RL

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Formulation

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